CM CASE VIGNETTES: Chronic Complications of Diabetes

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- No conflicts of interest to disclose

Learning Objectives

After participating in this session, attendees should be able to:

1. Recognize the clinical manifestations of diabetic microvascular complications including diabetic retinopathy, nephropathy and neuropathy
2. Explain methods to screen patients for chronic complications of diabetes mellitus
3. Identify physical exam or laboratory features that indicate underlying diabetes complications
4. Summarize common basic treatment options for diabetic complications

Question 1
Which of the following are considered diabetic complications?

A. DKA (diabetic ketoacidosis)
B. HHS (hyperglycemic hyperosmolar state)
C. Heart Disease
D. Stroke
E. Kidney Failure
F. Blindness
G. Amputation

Questions 1
Which of the following are considered diabetic complications?

A. **DKA (diabetic ketoacidosis)**
B. **HHS (hyperglycemic hyperosmolar state)**

ACUTE

C. **Heart Disease**
D. **Stroke**

CHRONIC/ MACROVASCULAR

E. **Kidney Failure**
F. **Blindness**
G. **Amputation**

CHRONIC/ MICROVASCULAR
(Today's focus)

If Blood Sugars are Controlled, this Dramatically reduces the Risk for **Microvascular Complications** (Based on large RCT)





DCCT- Type 1 Diabetics

Compared patients who intensively managed their glucose levels (A1c of 7.2) to those who did usual management (A1c of 9.2) :

Lower incidence of retinopathy, microalbuminuria, decreased progression of existing retinopathy and nephropathy, and a reduced incidence of neuropathy

UKPDS- Type 2 Diabetics

Compared A1C 7.0 % (intensive group) vs 7.9% (conventional group):

25 % risk reduction in microvascular disease in intensive group. No evidence of a threshold effect for A1C;

1% fall in A1C associated with 35% reduction in microvascular endpoints

Lancet 1998;352(9131):837-53. Lancet 1998 Sep 12;352(9131):854-65.

Question 2

A 19-year-old patient, previously well with no significant past medical history, presents to your office after hospital discharge 3 days ago. The patient was hospitalized with diabetic ketoacidosis and a brand new diagnosis of type 1 diabetes. The patient was treated and discharged on insulin.

What would you advise the patient regarding screening for eye and kidney complications?

A. Dilated eye exam and urine microalbumin now
B. Dilated eye exam in 6 months, urine microalbumin now
C. Dilated eye exam in 6 months, urine microalbumin in 6 months
D. Dilated eye exam in 5 years, urine microalbumin in 5 years

Microvascular Damage

- Small blood vessel/ nerve damage: retinopathy nephropathy, neuropathy
- Usually occurs over time
- Risk is higher if diabetes is not controlled (by A1c)

- Studies have shown that glycemic control does help to prevent microvascular complications

  (less clear evidence that glucose control prevents macrovascular disease)

Screening test

- Retinopathy- screen with a dilated eye exam or retinal camera to visualize back of eye

- Nephropathy- screen with a urine microalbumin to creatinine ratio normal <30 mg/g creatinine) (abbreviated UACR)





Remote reading or the use of USFDA–approved AI algorithms to improve access to diabetic retinopathy screening are appropriate screening strategies
Programs need to provide pathways for timely referral for a comprehensive eye examination when indicated

Retinopathy Screening

Because retinopathy is estimated to take at least 5 years to develop after the onset of hyperglycemia:

People with **type 1 diabetes** should have an initial dilated and comprehensive eye examination **5 years *after* the diagnosis of diabetes**

Retinopathy Screening

People with **type 2 diabetes**, who may have had years of undiagnosed diabetes and have a significant risk of prevalent diabetic retinopathy at the time of diagnosis:

Should have an initial dilated and comprehensive eye examination **at the time of diagnosis**

Monitoring

If there is no evidence of retinopathy from one or more annual eye exams and glycemic indicators are within the goal range, then screening every 1–2 years may be considered

If any level of diabetic retinopathy is present, subsequent dilated retinal examinations should be repeated at least annually

If retinopathy is progressing or sight-threatening, examinations required more frequently

Nephropathy Screening

At least annually, urinary albumin (e.g., spot urinary albumin-to-creatinine ratio [UACR]) and estimated glomerular filtration rate [eGFR] should be assessed:

- in people with **type 1 diabetes** with duration of **≥5 years**
- in all people with **type 2 diabetes** regardless of duration

Question 2

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B. Dilated eye exam in 6 months, urine microalbumin now

C. Dilated eye exam in 6 months, urine microalbumin in 6 months

D. Dilated eye exam in 5 years, urine microalbumin in 5 years *(screening for type 1 diabetes begins after disease duration of ≥5 years)*

Question 3

Which of the following are options to treat diabetic retinopathy? (more than 1 correct answer)

a. Anticoagulation
b. Corticosteroids
c. Intravitreous injections of anti–vascular endothelial growth factor (anti-VEGF)
d. Macular focal/grid laser photocoagulation
e. Panretinal laser photocoagulation

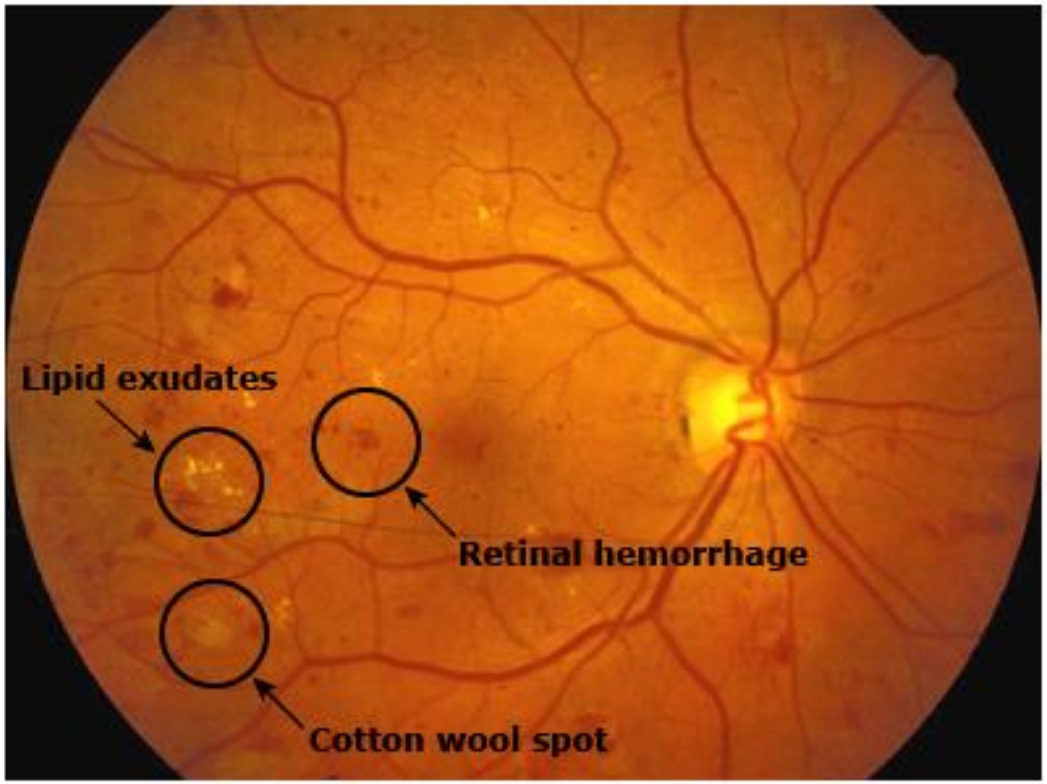
Retinopathy

Reduction in the progression of retinopathy is strongly related to the duration of diabetes and level of glucose control

Although glaucoma and cataracts may occur more frequently in patients with DM, when we screen for diabetic microvascular disease, we are looking for:

- Diabetes related macular edema
- Proliferative (PDR) and non proliferative retinopathy (NPDR)
- Neovascularization

Nonproliferative diabetic retinopathy: Appearance on fundus photograph



Color fundus photograph of nonproliferative diabetic retinopathy showing retinal (or "dot-blot") hemorrhages, yellow lipid (or "hard") exudates, and dull-white cotton wool spots (nerve fiber layer infarcts).

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Retinopathy

Screen to prevent loss of vision and to intervene with treatment when vision loss can be prevented or reversed

- Intravitreous injections of anti-VEGF is **first line in macular edema**
- Macular focal/grid or pan-retinal photocoagulation and / or intravitreal injections of corticosteroid can be used if persistent diabetic macular edema despite anti-VEGF, of if DME plus significant DR
- Laser photocoagulation can be used for DME and for microaneurysms as well as PDR

Question 3

Which of the following are options to treat diabetic retinopathy? (more than 1 correct answer)

a. Anticoagulation
b. **Corticosteroids *(not first line)***
c. **Intravitreous injections of anti–vascular endothelial growth factor (anti-VEGF)**
d. **Macular focal/grid laser photocoagulation**
e. **Panretinal laser photocoagulation**

Question 4

A 58 yr old patient with type 2 diabetes presents for a diabetes visit. A1c is 6.5%, Creatinine is 1.0mg/DL with GFR of 90. There was no retinopathy on dilated eye exam 2 months ago, and the urine microalbumin to creatinine ratio was normal (10). The patient inquires about prevention of future diabetic kidney disease. What would you recommend?

A. ACE inhibitor or ARB therapy
B. Aspirin
C. Calcium channel blocker
D. No additional therapy
E. Reduction in dietary protein

Nephropathy

Optimize glucose management (reduce risk /slow progression of CKD)

Optimize blood pressure (reduce risk /slow progression of CKD and reduce cardiovascular risk)

In (nonpregnant) people with T2DM **and CKD**

- SGLT2 inhibitors are recommended to reduce CKD progression and cardiovascular events in individuals with eGFR ≥20 mL/min/1.73 m^2 and urinary albumin ≥200 mg/g creatinine

In (nonpregnant) people with T2DM **and hypertension**

- Either an ACE inhibitor or an angiotensin receptor blocker (ARB) is recommended for those with albuminuria (UACR 30–299 mg/g creatinine), strongly recommended for severely increased albuminuria (UACR ≥300 mg/g creatinine) and/or eGFR <60 mL/min/1.73 m^2 to prevent progression of kidney disease and reduce cardiovascular events

Prevention of Nephropathy

An ACE inhibitor or an ARB is **not recommended** for the primary prevention of CKD in people with diabetes who have

- normal blood pressure
- normal UACR (<30 mg/g creatinine)
- normal eGFR

Question 4

A 58 yr old patient with type 2 diabetes presents for a diabetes visit. A1c is 6.5%, Creatinine is 1.0mg/DL with GFR of 90. There was no retinopathy on dilated eye exam 2 months ago, and the urine microalbumin to creatinine ratio was normal (10). The patient inquires about prevention of future diabetic kidney disease. What would you recommend?

A. ACE inhibitor or ARB therapy *(patient does not have hypertension, microalbuminuria, or reduced GFR)*
B. Aspirin *(no evidence)*
C. Calcium channel blocker *(no evidence- patient without hypertension)*
D. No additional therapy
E. Reduction in dietary protein *(no evidence, patent has normal GFR)*

Question 5

A 64 year old patient with type 2 diabetes since the age of 32 and obesity presents to the office for annual exam. Diabetes control has been poor, most recent A1c is 10.6%.

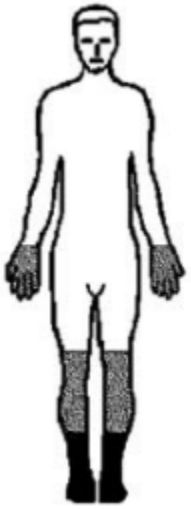
On exam you note a shallow ulcer on the bottom of the right foot. Which of the following would you also expect on the physical exam?

A. Hyperreflexia
B. Increased sensitivity to pinprick
C. Increased sensitivity to vibration
D. Increased sensitivity to temperature
E. Reduced sensation to monofilament

Diabetic Neuropathies

- Distal symmetric polyneuropathy
- Autonomic neuropathy
- Thoracic and lumbar nerve root disease, causing polyradiculopathies
- Individual cranial and peripheral nerve involvement causing focal mononeuropathies (ex CN III)
- Asymmetric involvement of multiple peripheral nerves, resulting in a mononeuropathy multiplex

Peripheral Neuropathy



Damage to nerve and/or small vasculature to the nerves

- Burning / tingling pain sensations (typically in the “stocking glove distribution”)
- Loss of protective sensation (numbness)
- Loss of temperature/ pinprick sensation
- Loss of reflexes, vibratory sensation

Foot Care

- Foot ulcerations and amputations are common complications
  - May be consequences of several factors: peripheral neuropathy, peripheral arterial disease (PAD), and foot deformities
  - Early recognition of at-risk feet, prompt treatment of ulcerations and other lower extremity complications can delay or prevent adverse outcome
- Risk Factors: prior history of ulceration, amputation, foot deformity, angioplasty or vascular surgery, cigarette smoking, retinopathy, renal disease
- Consider Screening for peripheral arterial disease (PAD)

Foot Care

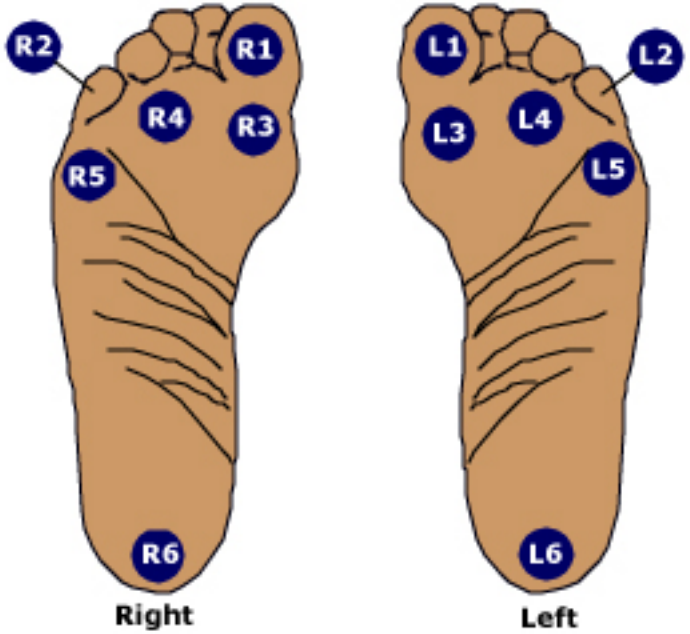
Perform a comprehensive foot evaluation **at least annually**
Individuals with evidence of sensory loss or prior ulceration or amputation should have their feet inspected at every visit

Inspection of skin, for foot deformities, vascular assessment (pulses)

Neurological assessment – test for loss of protective sensation

- 10-g monofilament testing with at least one other test: pinprick, temperature, or vibration),
- Vibration- 128-Hz tuning fork, to the bony prominence at the dorsum of the first toe, proximal to the nail bed. Ask the patient to report perception of start of vibration and cessation
- Pinprick sensation
- Ipswich touch test- quicker and simpler method to assess sensation, designed to promote more widespread diabetic foot examinations (lightly, for 1-2 seconds) touch tips of the first, third, and fifth toes with index finger. Reduced sensation is defined as ≥2 of 6 insensate areas (counting both feet)

Testing sites for sensation in evaluation of diabetic foot



The monofilament used to evaluate sensation should be tested at each of the 12 sites shown, which represent the most common sites of ulcer formation. Failure to detect cutaneous pressure at any site indicates that the patient is at high risk for future ulceration.

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American Diabetes Association®

From: **Comprehensive Foot Examination and Risk Assessment, A report of the Task Force of the Foot Care Interest Group of the American Diabetes Association, with endorsement by the American Association of Clinical Endocrinologists**

Diabetes Care. 2008;31(8):1679-1685. doi:10.2337/dc08-9021

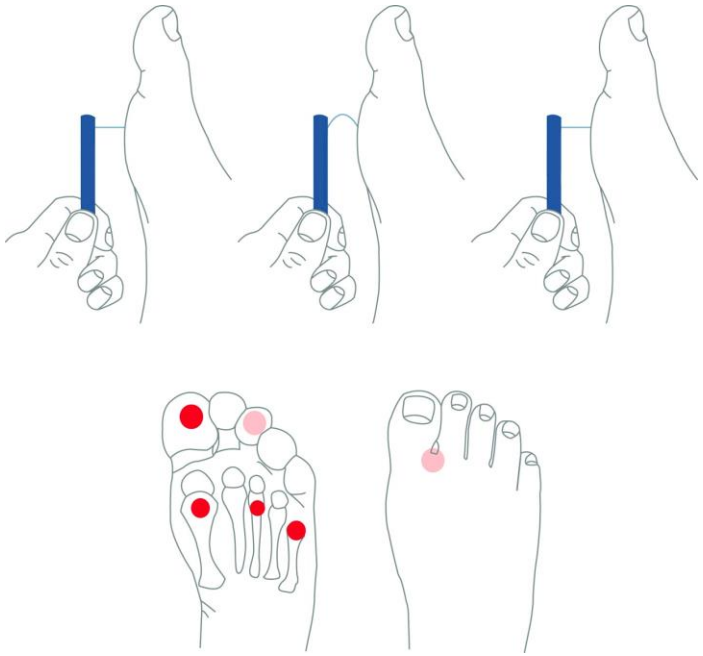


Figure Legend:

Upper panel: For performance of the 10-g monofilament test, the device is placed perpendicular to the skin, with pressure applied until the monofilament buckles. It should be held in place for ~1 s and then released. Lower panel: The monofilament test should be performed at the highlighted sites while the patient's eyes are closed.

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Foot Care

Provide general preventive foot self-care education to all people with diabetes

Avoid smoking, walking barefoot, wearing tight or poorly fitting shoes

Self inspection: palpation or visual inspection (with an unbreakable mirror) for daily surveillance of early foot problems

Neuropathy Treatment

Optimize glucose management to prevent or delay development of neuropathy in people with T1DM; also shown slow the progression of neuropathy in T2DM

Optimize blood pressure and lipids to reduce the risk or slow the progression of diabetic neuropathy

Assess and treat pain related to diabetic peripheral neuropathy and symptoms of autonomic neuropathy

- Gabapentinoids, serotonin-norepinephrine reuptake inhibitors, tricyclic antidepressants, and sodium channel blockers are recommended as initial pharmacologic treatments for neuropathic pain in diabetes

Refer to neurologist or pain specialist when adequate pain management is not achieved within the scope of practice of the treating clinician

A Note on Neuropathy

Diabetic neuropathy is a diagnosis of exclusion. Nondiabetic neuropathies may be present in people with diabetes and may be treatable

Yes, Diabetes is a possible cause…but please don't overlook other possible causes

- Toxins
- Alcohol use
- B12 deficiency
- Vasculitis
- Demyelinating polyneuropathies
- Malignancies

Question 5

A 64 year old patient with type 2 diabetes since the age of 32 and obesity presents to the office for annual exam. Diabetes control has been poor, most recent A1c is 10.6%.

On exam you note a shallow ulcer on the bottom of the right foot. Which of the following would you also expect on the physical exam?

A. Hyperreflexia *(hypo/ reduced)*
B. Increased sensitivity to pinprick *(reduced)*
C. Increased sensitivity to vibration *(reduced)*
D. Increased sensitivity to temperature *(reduced)*
E. Reduced sensation to monofilament

Question 6

At a clinic, you see a 49 year old male who has had type 1 diabetes since the age of 4. Recent A1c is 9.2%. He is concerned about erectile dysfunction. On physical exam, Vitals signs show normal temperature, blood pressure and respiration, and heart rate is 110. You have noted that his heart rates have been elevated for his last 3 office visits.

What is the most likely cause for tachycardia in this patient?

A. Caffeine
B. Chronic infection
C. Nephropathy
D. Neuropathy
E. Pulmonary embolus

Diabetic Autonomic Neuropathy

Type 1 diabetes for ≥5 years and all individuals with type 2 diabetes

Assess annually for autonomic neuropathy

Ask symptoms and signs- history and physical examination

Symptoms- Autonomic Neuropathy

Orthostatic intolerance (dizziness, lightheadedness, or weakness with standing)

Syncope

Exercise intolerance

Constipation, diarrhea

Urinary retention, urinary incontinence

Changes in sweat function

Clinical Manifestations- Autonomic Neuropathy

Resting tachycardia

Orthostatic hypotension

Gastroparesis

Fecal incontinence

Erectile dysfunction

Neurogenic bladder

Sudomotor dysfunction with either increased or decreased sweating

Autonomic Neuropathy, Treatment

- Prevention- with glucose control
- Obtain glycemic control slowly
- Address modifiable risk factors (hypertension, hyperlipidemia, and smoking)
- Treatments depend on the type- most is supportive

  Ex. Cardiac- remove meds that can worsen orthostatic HTN, increase fluids, falls precautions

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What is the most likely cause for tachycardia in this patient?

A. Caffeine
B. Chronic infection *(no fever, less likely)*
C. Nephropathy
D. Neuropathy (autonomic)
E. Pulmonary embolus *(no other signs, this has been chronic)*

Question 7

A 45-year-old woman with a 20-year history of type 1 diabetes presents complaining of early satiety, bloating, and intermittent nausea and vomiting for the past 6 months. Her A1c is 8.6%. She reports that her glucose levels are difficult to manage, with frequent postprandial hyperglycemia and occasional hypoglycemia hours after eating. She denies any recent travel, fever, or changes in diet. On physical exam, there is mild epigastric tenderness without guarding or rebound. Her labs are unremarkable, and an abdominal ultrasound was normal

What is the most likely explanation for her symptoms?

A. Exocrine pancreatic insufficiency
B. Peptic ulcer disease
C. Vagus neuropathy
D. Cholelithiasis
E. Celiac disease

Autonomic Neuropathy

Longstanding type 1 diabetes and poor glycemic control – symptoms consistent with gastroparesis

Classic symptoms of delayed gastric emptying

- early satiety
- bloating
- nausea vomiting,
- erratic glycemic patterns

Gastroparesis

Cumulative incidence of gastroparesis over 10 years in patients with type 1 (5%) and type 2 DM (1%) per population studies (0.2% in controls)

Not all gastroparesis is from diabetes

Other causes: medications, viruses, neurologic disorders, multiple sclerosis, brainstem stroke or tumor, parkinsonism, amyloid neuropathy, paraneoplastic syndrome/ autoimmune, are some examples.

Gastroparesis

Diagnosis is clinical but may be confirmed with gastric emptying studies

Management includes:

- Dietary modifications (small, frequent meals)
- Glucose optimization
- Sometimes prokinetic agents (Metoclopramide)

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QUESTIONS?

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